DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW

Study and Evaluation Scheme • B.Tech Computer Science and Engineering

S.No.	Course Code	Course Title	Periods			Evaluation Scheme		Total Marks	Credit
			L	T	P	Internal (CT+TA)	ESE		
1	KCS301	Data Structures	3	1	0	30 + 20	100	150	4
2	KCS302	Computer Organization and Architecture	3	1	0	30 + 20	100	150	4

KCS301: DATA STRUCTURES

Semester: III • L-T-P: 3-1-0 • Credits: 4

Course Outcomes: At the end of course, students will be able to analyze algorithm efficiency and choose appropriate non-linear data structures for complex problems.

UNIT I: INTRODUCTION, ARRAYS & LINKED LISTS

• Introduction: Basic Terminology, Elementary Data Organization, Built-in Data Types in C

• Algorithm Analysis: Efficiency of Algorithms, Time and Space Complexity, Asymptotic Notations (Big O, Omega, Theta), Time-Space Tradeoff

• Arrays: Single and Multidimensional Arrays, Row-Major and Column-Major Representations, Applications of arrays, Sparse Matrices

• Linked Lists: Singly Linked Lists, Doubly Linked Lists, Circular Linked Lists, Operations on Linked Lists (Insertion, Deletion, Traversal), Polynomial Arithmetic

UNIT II: STACKS AND QUEUES

• Stacks: Primitive Stack Operations (Push & Pop), Array and Linked Implementation of Stacks in C

• Applications of Stacks: Infix to Postfix Conversion, Evaluation of Postfix Expressions, Recursion Principles, Tail Recursion

• Queues: Array and Linked Implementations, Circular Queues, Dequeues, Priority Queues

UNIT III: SEARCHING, SORTING & HASHING

• Searching: Sequential Search, Binary Search, Indexed Sequential Search

• Hashing: Hash Functions, Collision Resolution Techniques, Open Addressing, Chaining

• Sorting Algorithms: Insertion Sort, Selection Sort, Bubble Sort, Quick Sort, Merge Sort, Heap Sort, Radix Sort

UNIT IV: GRAPHS

• Graph Terminology: Directed and Undirected Graphs, Subgraphs, Connected Components, Degree

• Graph Representations: Adjacency Matrix, Adjacency List

• Graph Traversals: Depth First Search (DFS), Breadth First Search (BFS)

• Minimum Spanning Trees & Shortest Paths: Prim's Algorithm, Kruskal's Algorithm, Dijkstra's Algorithm, Warshall's Algorithm

UNIT V: TREES

• Tree Terminology: Binary Trees, Binary Search Trees (BST), Complete Binary Trees

• Tree Traversals: Inorder, Preorder, Postorder Traversals, Construction from Traversals

• BST Operations: Insertion, Deletion, Search in Binary Search Tree

• Advanced Trees: AVL Trees, B-Trees, Huffman Coding, Binary Heaps

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Suggested Text Books & References:

1. Aaron M. Tenenbaum, "Data Structures Using C", Pearson Education.
2. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publications.

Examination Note: The question paper will consist of Section A (Short questions), Section B (Medium questions), and Section C (Long analytical questions).